



Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Mathematics (8MA0)

Paper 21 Statistics

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General Comments

Overall, the paper was accessible to learners, with little evidence to suggest that learners had insufficient time to complete it. Although the paper contained four questions, each question was clearly structured and provided opportunities for learners to continue to make progress after earlier errors.

Question 1 provided an accessible start to the paper and allowed many learners to engage confidently with the assessment. Questions 2 and 3 maintained a similar level of demand, and many responses showed that learners were able to make progress through these questions.

Question 4 also began with familiar content, before becoming more demanding in the later parts. As a result, the paper gave learners opportunities to demonstrate their understanding across a range of levels. The most discriminating features were seen towards the end of the paper, particularly in the accuracy of calculations and the precision of written responses.

Learners should be reminded of the importance of using the context of the question clearly in written responses. In some responses, key terms were missing or explanations were too imprecise to gain full credit.

Comments on Individual Questions

Question 1

As well as assessing learners' ability to tackle histogram questions, this question assessed learners' ability using statistical representations and recognising information from the Large Data Set.

Part (a)

Most learners performed well on part (a) and recalled 'Total Daily Rainfall' would be measured in millimetres. Incorrect answers seen were often relating to volume e.g. mm^3 or litres or mm/day .

Part (b)

In part (b), approaches using frequency density were the most successful, and correct, concise solutions were often seen. Many learners recognised that frequency is proportional to area. In proportionality approaches, however, some responses showed uncertainty about whether to multiply or divide by the value of k .

Some solutions included several calculations with limited linking commentary. Including clearer explanation between steps would help to show the method being used, particularly where an arithmetic error is made. When a counting squares approach was used, errors in counting were sometimes seen. A common source of incorrect answers was treating the first two classes as having a combined frequency of 22, rather than using 22 for the second class only.

Part (c)

Most learners were successful with part (c)(i) and recognised that tr is a value between 0 and 0.05 in the Large Data Set, with most learners opting to replace tr with 0. Learners were less successful when comparing their tr value with the mean, with very few learners explaining that adding a value lower than the mean to a set of data will have the effect of lowering the mean. Many learners were able to reason that by adding a data point of zero it would cause the sum of the rainfall to remain the same, but you would now be dividing by a larger frequency.

Learners should be reminded to read the wording of the question carefully and to provide a sufficiently detailed explanation of their reasoning, making clear use of the context where required.

Question 2

Part (a)

Most learners were successful in part (a), with few learners adding another set of branches onto L' and not gaining full marks.

Part (b)

In part (b), many responses did not gain the mark for finding the probability that a learner studied both a science **and** a humanities subject. This required learners to recognise that, since exactly two of the three subject options were chosen, studying both science and humanities was equivalent to not studying a language. The required value could therefore be read directly from the completed diagram.

A common incorrect answer was 0. This often arose where responses did not make use of the initial information that exactly two of the three subject options had been chosen.

Part (c)

In part (c), fewer responses gained full credit. A common incorrect answer was 0.14, particularly where the answer to part (b) had been given as 0. Many responses gave the correct hypotheses and stated the required Binomial model for the question.

Where errors occurred, these often involved using an incorrect probability. A common approach was to use $p = 0.25$, derived from dividing 8 by 32. Many responses derived the correct probability needed for the concluding statement, but some gave this probability to 2 significant figures rather than at least 3 significant figures, so the accuracy mark was not awarded. The critical region approach was seen less often, but where it was used, it was usually applied correctly.

The conclusion was usually given in context and was generally correct although some did not mention Sam if they were commenting on the correctness of the claim and others said the number rather than proportion had not decreased.

Question 3

Many learners performed well on this question and could score throughout the question.

Part (a)

In part (a), many responses correctly identified that the y -coordinate of P was an outlier. Successful responses often used inequality notation effectively to compare values and justify the conclusion. Some straightforward errors were seen, so learners should be reminded to show the formula with the values substituted in before carrying out any calculations.

Part (b)

Parts (b) and (c) were well answered. Many responses interpreted the positive correlation in context in part (b), rather than only stating that a positive correlation was shown.

Part (c)

In part (c), some responses used the decimal equivalent of 45% when substituting into the equation, which led to an incorrect answer. Learners should be reminded that, in this context, the percentage value should be substituted as 45 rather than 0.45. A common incorrect answer was 59%, arising from this substitution.

Many responses gave 70% rather than 72.5%, indicating that the estimate had been rounded to 1 significant figure. Another common incorrect answer was 72%, where the value had been truncated rather than rounded appropriately. Learners should be reminded to consider the required level of accuracy when giving an estimated percentage in context.

Part (d)

Part (d) was more challenging. Many responses gave a general statement, such as “0.3 is the gradient of the regression line”, rather than interpreting the value in the context of the question. To gain full credit, responses needed to explain the meaning of the value clearly in context, including reference to an “increase” and to “turnout”.

Question 4

This question proved to be challenging, with many responses making no attempt at all.

Part (a)

In part (a), values of x were often substituted into the expression. To gain full credit, responses needed to show the three probabilities being summed and the total set equal to 1. The verification approach was seen less often. Learners should be reminded to use a complete and clearly presented method when forming and solving this type of equation.

Part (b)

In part (b), many responses interpreted the word “value” as meaning “probability”, rather than the value of X . This led to differences in the probabilities being calculated. Some responses included negative values, rather than recognising that the difference must be positive, giving more than four possible values in total.

The difference 0 was sometimes omitted, where responses treated X_1 and X_2 as needing to be different. Even when the correct set of values for D was not listed, many responses correctly calculated $P(D = 7)$ using X values of 3 and 4.

Part (c)

In part (c), solving the quadratic inequality was a challenging aspect of the question. Some responses gave $D = 6$ and $D = 2$ as solutions, or gave inequalities such as $D > 6$ or $D > 2$. Even where the inequality was solved correctly, further progress depended on having identified suitable values of D in part (b).

Some responses used a binomial distribution at this stage, rather than working from the possible values of D . The substitution approach was often more successful when a reasonably complete set of values for D had been listed in part (b), and these responses generally made clear which values of D had been selected.

Clear labelling was important in this question. Correct values for $P(D = 9)$ or $P(D = 2)$ could not always be credited unless the final answer was correct. Using $1 - P(D = 2)$ was the most efficient final step, and this approach was sometimes seen.